Investigation of Compressible Flow Formulation in the Low- Mach, Low-Reynolds Regime (2)

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Objective:

The objective of this study is to investigate whether the compressible flow equations implemented in ISAAC can accurately predict flow behaviour at low Reynolds numbers (Re = 10,000 and below). In such regimes, the corresponding Mach numbers become very small, approaching the incompressible limit. Many numerical solvers designed for compressible flows tend to lose accuracy or stability under these conditions. Therefore, this study aims to assess the applicability and limitations of the compressible formulation in ISAAC for low-Reynolds-number flows.

Nomenclature:

ρ:	Density	L:	Characteristic length
μ:	Dynamic viscosity	M:	Mach number
C_L:	Left coefficient	V:	Velocity
C_D:	Drag coefficient	Re:	Reynolds number

Introduction:

At low Mach numbers, the large disparity between acoustic and convective wave speeds leads to stiffness and poor conditioning of the compressible Navier–Stokes equations, resulting in degraded convergence and potential inaccuracies (Turkel, 1999). This behaviour poses a significant challenge for numerical solvers that are primarily designed for compressible flow regimes.

However, an important question arises regarding the limits of applicability of these formulations: at what point do compressible solvers cease to provide reliable results as the Mach number decreases? For low-Reynolds-number flows—where velocities are typically small and the flow approaches the incompressible limit—it becomes essential to evaluate whether compressible formulations remain accurate and efficient, or whether alternative approaches are required.

To investigate these limitations, a specific test case is considered. The flow around an aerofoil at low Reynolds numbers is analysed, and the corresponding flow properties are determined.

As an upper bound for the present study, a Reynolds number of $Re = 10{,}000$is selected. At this limit, it is first necessary to evaluate the corresponding Mach number to assess the relevance of compressibility effects.

Given:

$Re = 10{,}000$

$L = 1m$

$V = 1\ m/s$

$\rho = 1\ kg/m^3$

It can be shown that, using the Reynolds number definition:

$\mu = 0.0001\ kg/m^3$

With such variables Mach number will be:

$$M = 0.0029$$

Note: For more details on the calculation procedures please visit:

Investigation_of_Compressible_Flow_Formulation_in_the_Low_Mach_Low_Reynolds_Regime.pdf

Mesh:

To generate the mesh from Pointwise, export the grid in **Plot3D** format. Before exporting, ensure that the following settings are correctly configured:

- **Precision**: Double precision
- **Grid Type**: Multi-block

Note: The mesh has 601 × 301 nodes, ISAAC is an older-style Fortran code, it uses fixed-size arrays, not dynamic memory. The code was compiled for a slightly smaller case, with such high number of nodes, two error messages will appear:

"MAIN : Storage required for RES 922625 exceeds allocated 918080 => Increase MXSIZE in MAIN and re-submit.

MAIN : Storage required for cross-section 184525 exceeds allocated 183616 => Increase MXSECT in MAIN and re-submit."

To increase the allocated array size of variables:

1. open the file main.F which is located at:

   ISAAC/src/main/main.F

2. increase the two variables MXSIZE and MXSECT to:

   PARAMETER (MXSIZE = 1200000)

   PARAMETER (MXSECT = 250000)

Note: The values were increased beyond the required size indicated in the error message to accommodate future calculations.

Boundary Conditions:

Figure (1) illustrates the orientation of the *i* and *j* indices. The *j*-index starts at the outer boundaries (top and bottom) and progresses toward the branch cut (the central line). The *i*-index runs from the far-right boundary to the corresponding point on the lower part of the domain.

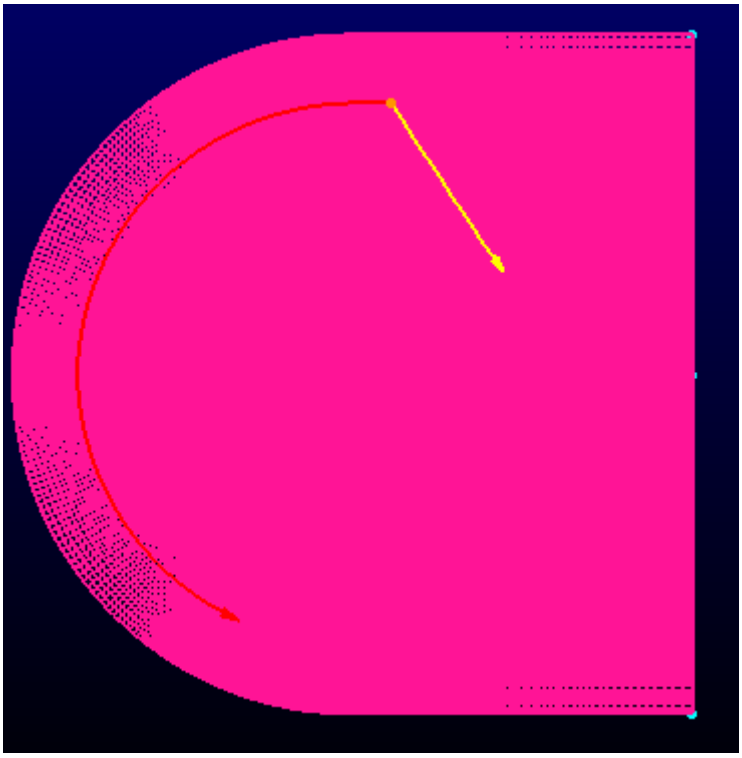

Figure 1: In yellow the orientation of the j index and in red the orientation of the i index

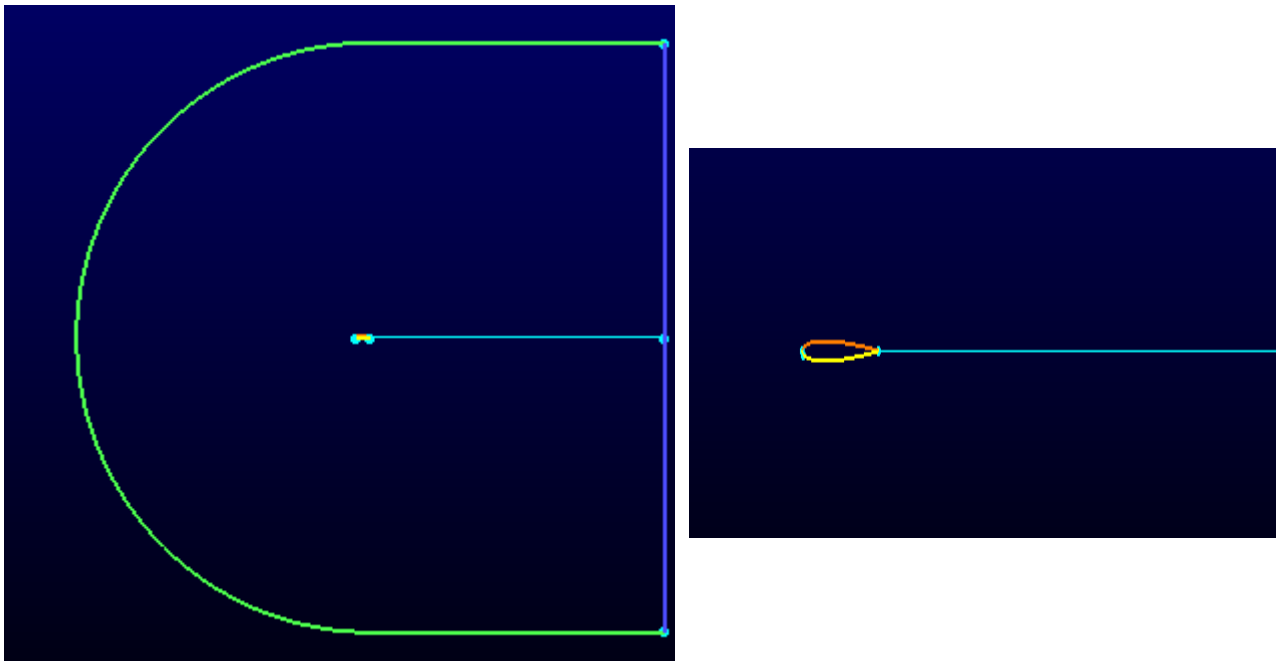

Figure 2: The right side shows a zoomed-out perspective of the domain, whereas the left side presents a zoomed-in view.

The following section explains how to define boundary conditions in ISAAC in a manner consistent with the index orientation shown in Figure (1).

- The first line specifies the total number of boundary conditions applied

  'BC' 8

- Inlet (Figure 2, left – shown in green)

  'BC' 1 1 1 601 1 2 'FARFIELD'

  This defines the inlet boundary using a Farfield condition.

- Aerofoil upper surface (shown in orange)

  'BC' 201 301 1 301 301 2 'WALL'

  Defined as a solid wall.

- Aerofoil lower surface (shown in yellow)

  'BC' 301 301 1 201 301 2 'WALL'

  Also defined as a wall.

- Upper outlet boundary (upper part of the purple line)

  'BC' 1 1 1 1 301 2 'EXTRAPOLATE'

  This represents the upper portion of the outlet boundary, where flow variables are extrapolated from the interior.

- Lower outlet boundary (lower part of the purple line)

  'BC' 601 1 1 601 301 2 'EXTRAPOLATE'

  This defines the lower portion of the outlet boundary, also using extrapolation.

- Branch cut / wake line (middle line including aerofoil junction)

  'BC' 1 301 1 601 301 2 'EXTRAPOLATE'

  This boundary corresponds to the branch cut (the central line extending from the aerofoil), where flow variables are extrapolated to maintain continuity across the wake.

- Front and back planes of the 2D mesh

  'BC' 1 1 1 601 301 1 'EXTRAPOLATE'

  'BC' 1 1 2 601 301 2 'EXTRAPOLATE'

  These boundaries correspond to the front and back planes in the *k*-direction. The use of extrapolation enforces no variation in the *k*-direction, thereby ensuring a two-dimensional simulation.

Setup:

The code requires the specification of the Reynolds number (10,000 in this case) and the Mach number (0.0029).

Results:

From Figure (3), the solver fails to reproduce the expected physical behaviour, as the lift coefficient $C_L$clearly diverges to values on the order of $O(5)$.

```
ITER =     491     491 RES =  2.1887846E-05   3.2650441E-06   3.0136320E-06   0.0000000E+00   5.4723044E-05   5.9105261E-05
Iter =     491     491 CL = -1.698639E+05 CD =  5.627058E+00
MAIN  : DT = 1.000E-06  CFLMIN = 7.750E-06 At BLOCK:I,J,K=(  1: 276,   2,   2)
                        CFLMAX = 1.044E-03 At BLOCK:I,J,K=(  1: 301, 301,   2)
ITER =     492     492 RES =  2.1881330E-05   3.2708278E-06   3.0189645E-06   0.0000000E+00   5.4706758E-05   5.9088361E-05
Iter =     492     492 CL = -1.698639E+05 CD =  5.637196E+00
MAIN  : DT = 1.000E-06  CFLMIN = 7.750E-06 At BLOCK:I,J,K=(  1: 276,   2,   2)
                        CFLMAX = 1.044E-03 At BLOCK:I,J,K=(  1: 301, 301,   2)
ITER =     493     493 RES =  2.1874817E-05   3.2766079E-06   3.0242940E-06   0.0000000E+00   5.4690479E-05   5.9071470E-05
Iter =     493     493 CL = -1.698639E+05 CD =  5.647332E+00
MAIN  : DT = 1.000E-06  CFLMIN = 7.750E-06 At BLOCK:I,J,K=(  1: 276,   2,   2)
                        CFLMAX = 1.044E-03 At BLOCK:I,J,K=(  1: 301, 301,   2)
ITER =     494     494 RES =  2.1868306E-05   3.2823846E-06   3.0296206E-06   0.0000000E+00   5.4674207E-05   5.9054588E-05
Iter =     494     494 CL = -1.698639E+05 CD =  5.657464E+00
MAIN  : DT = 1.000E-06  CFLMIN = 7.750E-06 At BLOCK:I,J,K=(  1: 276,   2,   2)
                        CFLMAX = 1.044E-03 At BLOCK:I,J,K=(  1: 301, 301,   2)
ITER =     495     495 RES =  2.1861798E-05   3.2881578E-06   3.0349443E-06   0.0000000E+00   5.4657942E-05   5.9037714E-05
Iter =     495     495 CL = -1.698639E+05 CD =  5.667592E+00
MAIN  : DT = 1.000E-06  CFLMIN = 7.750E-06 At BLOCK:I,J,K=(  1: 276,   2,   2)
                        CFLMAX = 1.044E-03 At BLOCK:I,J,K=(  1: 301, 301,   2)
ITER =     496     496 RES =  2.1855294E-05   3.2939275E-06   3.0402651E-06   0.0000000E+00   5.4641684E-05   5.9020850E-05
Iter =     496     496 CL = -1.698639E+05 CD =  5.677718E+00
MAIN  : DT = 1.000E-06  CFLMIN = 7.750E-06 At BLOCK:I,J,K=(  1: 276,   2,   2)
                        CFLMAX = 1.044E-03 At BLOCK:I,J,K=(  1: 301, 301,   2)
ITER =     497     497 RES =  2.1848792E-05   3.2996937E-06   3.0455831E-06   0.0000000E+00   5.4625434E-05   5.9003994E-05
Iter =     497     497 CL = -1.698639E+05 CD =  5.687841E+00
MAIN  : DT = 1.000E-06  CFLMIN = 7.750E-06 At BLOCK:I,J,K=(  1: 276,   2,   2)
                        CFLMAX = 1.044E-03 At BLOCK:I,J,K=(  1: 301, 301,   2)
ITER =     498     498 RES =  2.1842293E-05   3.3054565E-06   3.0508981E-06   0.0000000E+00   5.4609191E-05   5.8987147E-05
Iter =     498     498 CL = -1.698639E+05 CD =  5.697960E+00
MAIN  : DT = 1.000E-06  CFLMIN = 7.750E-06 At BLOCK:I,J,K=(  1: 276,   2,   2)
                        CFLMAX = 1.044E-03 At BLOCK:I,J,K=(  1: 301, 301,   2)
ITER =     499     499 RES =  2.1835797E-05   3.3112157E-06   3.0562102E-06   0.0000000E+00   5.4592954E-05   5.8970308E-05
Iter =     499     499 CL = -1.698639E+05 CD =  5.708076E+00
MAIN  : DT = 1.000E-06  CFLMIN = 7.750E-06 At BLOCK:I,J,K=(  1: 276,   2,   2)
                        CFLMAX = 1.044E-03 At BLOCK:I,J,K=(  1: 301, 301,   2)
ITER =     500     500 RES =  2.1829303E-05   3.3169714E-06   3.0615194E-06   0.0000000E+00   5.4576725E-05   5.8953479E-05
Iter =     500     500 CL = -1.698639E+05 CD =  5.718189E+00
```

Figure 3: Results output

Conclusion:

The observed divergence of the solver is attributed to the use of a compressible solver at very low Mach numbers. At this regimes, compressible solvers are known to suffer from numerical stiffness and inaccurate pressure–velocity coupling, which can lead to non-physical pressure fields and, consequently, deviations in the computed aerodynamic coefficients.

References:

- https://isaac-cfd.sourceforge.net/
- Turkel, E., 1999. Preconditioning techniques in computational fluid dynamics. Annual Review of Fluid Mechanics, 31, pp.385–416.